Week 11

SLAM, and Merge 1

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★ Merge 1

Electrical + Chassis

The claim

> *The vehicle steers to a commanded angle under firmware control, wheels off the ground.*

Closes bring-up **Stage 1**.

And it is the **E4 decision point**.

What must be shown

Closed-loop position tracking	**≤ 1° steady-state, no sustained oscillation**
Loop rate	≥ 200 Hz, measured
Limit behaviour	Commanded past ±22.5°, clamps rather than driving into the stop
Stall detection	Hold the wheel; the stall bit sets in `DbwStatus.faults`
Freewheel on power cut	Cut power; steering goes free, not locked
R/C signal-loss timeout	Stop the Teensy emitting pulses; the motor stops

E4 fires here, or not at all

If you are at 3° and have been adjusting gains for a week,
the honest answer is to invoke E4.

It costs you nothing — the decision record already calls it the planned action.

What **would** cost you is quietly continuing to tune while telling the standup it is nearly there.

The numbers Chassis owes everyone

Simulation and Software have both been waiting.

Merge 2 next week cannot happen without them.

For wheelbase, track width, wheel radius — show:

- the value
- **how it was measured**
- how much you trust it

SLAM

It is circular, and that is the difficulty

To build a map you need to know where you are.
To know where you are you need a map.

SLAM does both at once. It works because of **constraints**:
each scan constrains the pose that could have produced it,
given the map so far.

`slam_toolbox` maintains a **pose graph** — nodes are poses,
edges are scan-match constraints — and re-optimises as it goes.

Loop closure is where the value is

Drive down a corridor and back.

Odometry drifted, so the return does not land on the outward journey. The map shows the same wall **twice**, slightly apart.

Loop closure recognises a place already in the graph, adds an edge between two poses that were far apart, and re-optimises — distributing the accumulated error backwards over the whole loop.

Which is why the gate says 30 m

> `slam_toolbox` *closes a **≥ 30 m** hallway loop; repeated observations of the same wall agree within **≤ 10 cm**; the map loads in Nav2 without manual editing.*

A short straight run proves nothing.

Drift has not had room to accumulate, and **no loop has closed.**

Thresholds tighter than the defaults

```
resolution: 0.05
max_laser_range: 12.0
minimum_travel_distance: 0.2
minimum_travel_heading: 0.2
scan_buffer_size: 15
```

> **This vehicle moves at ≤ walking speed: waiting 0.5 m between scans would throw away most of the data on a slow indoor run.**

Inherit the defaults and the map is just **worse**,
with nothing to indicate why.

`base_link`, not `base_footprint`

B-MROVER's SLAM used `base_footprint` while its EKF used `base_link`.

MRider standardises on `base_link` **everywhere**.

There is no `base_footprint` anywhere in this stack —
if you see one, something is importing an unported config.

Transform ownership

Transform	Owner
`map → odom`	`slam_toolbox` — the drift correction
`odom → base_link`	EKF — the local pose estimate
`base_link → *`	`robot_state_publisher`

`map → odom` is not the robot's pose. It is the correction between where dead reckoning thinks it is and where SLAM believes it is.

When loop closure fires, that transform **jumps** — and that is correct.

Reading a bad map

Symptom	Usual cause
Doubled walls, worse after turning	Bad sensor extrinsic — Lab 4
Smeared walls	Timestamp / sync problem
Drifts steadily, never corrects	Loop never closed — no revisited place
Jumps and reshapes	Loop closure fired. Correct
Open space where a wall is	Glass, or above/below the scan plane
Robot walks through walls	TF broken, or two publishers of one transform

A bad map is evidence about the pipeline, not a verdict on SLAM.

Each symptom points at a different upstream component.

One more candidate, new this week

`mitt_dimensions.yaml` was **estimated** until today.

A wrong wheelbase produces a wrong yaw rate —
and therefore a **heading-dependent map error that looks exactly like a bad extrinsic.**

Put it at the top of your list until the measured values land.

Track work

Track	
Electrical	Stage 3 — relay MUX and E-stop; every failsafe row, **incl. E-stop with the laptop off**
Chassis	Mast and equipment plate; tip-over margin with the plate loaded
Simulation	Feed measured geometry into the twin; quantify the disagreement
Software	Nav2 re-derived for measured geometry — **both** radius parameters together

Merge 2 next week: Simulation + Software.